DL01 EX 07

1.a.

$$f(\mu,\sigma) = \underbrace{-\frac{(t-\mu)^2}{2\sigma^2}}_{①} \underbrace{-\log(\sqrt{2\pi}\cdot\sigma)}_{②}$$

① $\frac{\partial f}{\partial \mu}$:

$$\frac{\partial f}{\partial \mu} = -\frac{1}{2\sigma^2}\cdot 2(t-\mu)\cdot -1 - 0$$

$$= \frac{t-\mu}{\sigma^2}$$

② $\frac{\partial f}{\partial \sigma}$:

for $① = -\frac{(t-\mu)^2}{2}\cdot\sigma^{-2} \longrightarrow \frac{\partial ①}{\partial \sigma} = -\frac{(t-\mu)^2}{\cancel{2}}\cdot -\cancel{2}\cdot\sigma^{-3}$

$$= (t-\mu)^2\cdot\sigma^{-3}$$

$$= \frac{(t-\mu)^2}{\sigma^3}$$

for $② = -\log\sqrt{2\pi} - \log\sigma \rightarrow \frac{\partial ②}{\partial \sigma} = 0 - \frac{1}{\sigma} = -\frac{1}{\sigma}$

$$\Rightarrow \frac{\partial f}{\partial \sigma} = \frac{(t-\mu)^2}{\sigma^3} - \frac{1}{\sigma}$$

1.b.

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial \sigma} \cdot \frac{\partial \sigma}{\partial z}$$

① $\frac{\partial \sigma}{\partial z}$:

$\because \sigma(z) = \sqrt{1+z^2} \rightarrow$ define $1+z^2 = u$.

$$\sigma(z) = \sqrt{u} \rightarrow \frac{\partial \sigma}{\partial z} = \frac{\partial \sigma}{\partial u} \cdot \frac{\partial u}{\partial z} = \frac{1}{2} \cdot (1+z^2)^{-\frac{1}{2}} \cdot 2z$$

$$= \frac{1}{\cancel{2} \cdot \sqrt{1+z^2}} \cdot \cancel{2} \cdot z$$

$$= \frac{z}{\sqrt{1+z^2}}$$

$$= \frac{z}{\sigma}$$

② $\frac{\partial f}{\partial z} = \frac{\partial f}{\partial \sigma} \cdot \frac{\partial \sigma}{\partial z} = \left(\frac{(t-\mu)^2}{\sigma^3} - \frac{1}{\sigma}\right) \cdot \frac{z}{\sigma}$

$$= \left(\frac{(t-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\right) \cdot z$$

1. C

$$\left|\frac{\partial f}{\partial z}\right| = \left|\left(\frac{(t-\mu)^2}{\sigma^4} - \frac{1}{\sigma^2}\right)\cdot z\right|$$

① use triangle inequality : $|A-B| < |A| + |B|$

$$\left|\frac{\partial f}{\partial z}\right| \le \left(\left|\frac{(t-\mu)^2}{\sigma^4}\right| + \left|\frac{1}{\sigma^2}\right|\right)\cdot |z|$$

$$\le \left(\frac{(t-\mu)^2}{\sigma^4} + \frac{1}{\sigma^2}\right)\cdot |z|$$

② we know $\sigma = \sqrt{1+z^2} \ge 1 \longrightarrow \sigma \ge \sqrt{z^2} = |z|$

$$\rightarrow \left|\frac{\partial f}{\partial z}\right| \le \frac{(t-\mu)^2}{\sigma^4}\cdot |z| + \frac{1}{\sigma^2}\cdot |z|$$

$$= \frac{(t-\mu)^2}{\sigma^3}\cdot \underbrace{\frac{|z|}{\sigma}}_{\le 1} + \frac{|z|}{\sigma}\cdot\frac{1}{\sigma}$$

$$\le \frac{(t-\mu)^2}{\sigma^3} + \frac{1}{\sigma}$$

we know $\sigma \ge 1$

if μ close to t, no gradient explosion

$$\le \frac{(t-\mu)^2}{\sigma^2} + 1$$

$$\le \frac{(t-\mu)^2}{1} + 1 = \boxed{(t-\mu)^2 + 1}$$

1.d

① $\frac{\partial J}{\partial \theta} = \frac{1}{N} \cdot \sum_{i=1}^{N} \frac{\partial J_i}{\partial \theta_i}$

② $\frac{\partial J_i}{\partial \theta_i} = \frac{\partial J_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \theta} + \frac{\partial J_i}{\partial z_i} * \frac{\partial z_i}{\partial \theta}$

③ ①+② : $\boxed{\frac{\partial J}{\partial \theta} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{\partial J_i}{\partial \mu_i} \cdot \frac{\partial \mu_i}{\partial \theta} + \frac{\partial J_i}{\partial z_i} \cdot \frac{\partial z_i}{\partial \theta} \right)}$

1.e.

① for $\frac{\partial J}{\partial w}$, we need $\frac{\partial \mu_i}{\partial w}$ and $\frac{\partial z_i}{\partial w}$

1) $\mu_i = w^T \cdot x_i \rightarrow \frac{\partial \mu_i}{\partial w} = x_i$

2) $z_i = v^T \cdot x_i \longrightarrow \frac{\partial z_i}{\partial w} = 0$

$\Rightarrow \frac{\partial J}{\partial w} = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{\partial J_i}{\partial \mu_i} \cdot x_i + 0 \right) = \frac{1}{N} \sum_{i=1}^{N} \left(\frac{\partial J_i}{\partial \mu_i} \cdot x_i \right)$

$= \boxed{\frac{1}{N} \sum_{i=1}^{N} \frac{t_i - \mu}{\sigma^2} \cdot x_i}$

② for $\frac{\partial J}{\partial v}$, we need $\frac{\partial \mu_i}{\partial v}$ and $\frac{\partial z_i}{\partial v}$

1) $\frac{\partial \mu_i}{\partial v} = 0$ 2) $\frac{\partial z_i}{\partial v} = x_i$

3) $\Rightarrow \frac{\partial J}{\partial v} = \frac{1}{N} \sum_{i=1}^{N} \left(0 + \frac{\partial J_i}{\partial z_i} \cdot x_i \right) = \boxed{\frac{1}{N} \sum_{i=1}^{N} \left(\frac{(t_i - \mu_i)^2}{\sigma_i^4} - \frac{1}{\sigma_i^2} \right) \cdot z_i \cdot x_i}$